



AI-Powered Smart Karyotyping & FISH Analysis System

Transforming Cytogenetic Analysis with Advanced Artificial Intelligence

Key Highlights

- AI-Powered Chromosome Segmentation & Classification (> 90% validated Accuracy)
- < 2 Minutes Analysis Time
- G-Banding & FISH Support
- Wide Range of Microscopes & Imaging Systems
- 100% guaranteed data security
- Best price in the market

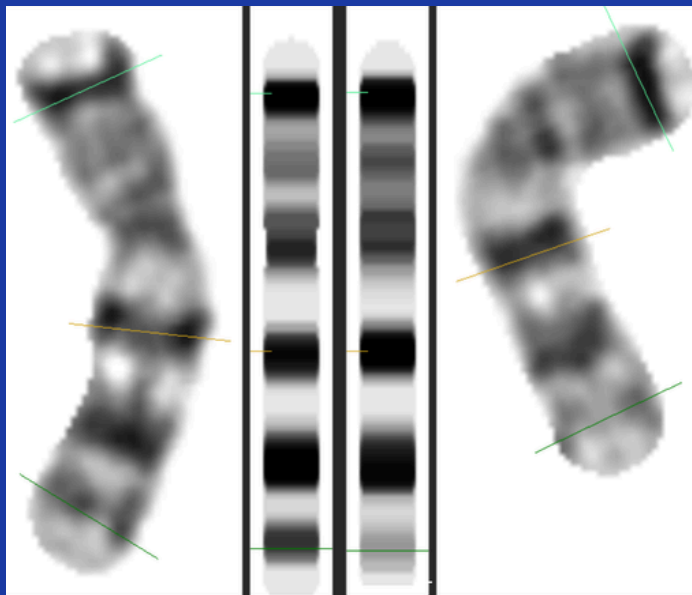
What is ChromoTraQ?

ChromoTraQ is an AI-powered decision support system designed to assist cytogenetic professionals in accurate and efficient genetic analysis.

It supports chromosome segmentation, classification, and abnormality detection to streamline laboratory workflows and enhance diagnostic confidence.

Developed by Sri Lankan Software Company Quantine Technologies (Pvt) Ltd. under the academic guidance of the Human Genetics Unit, Faculty of Medicine, University of Colombo, Sri Lanka.

Comparing Band Patterns in a Pair



Device Compatibility & Minimum Requirements

ChromoTraQ is designed for flexibility and broad compatibility.

Camera :

- Industrial CMOS or CCD camera ≥ 2 MP resolution

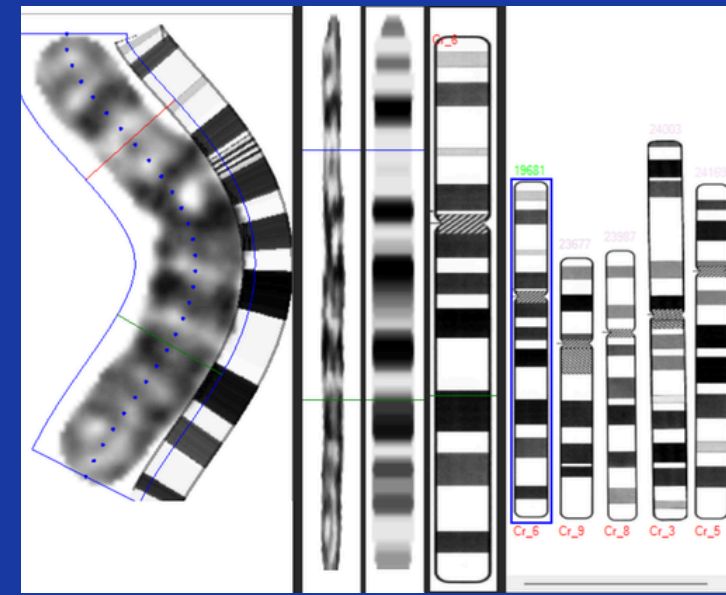
Microscope :

- Trinocular biological microscope
- 10 $\times$ objective (NA ≥ 0.25)
- 100 $\times$ oil immersion objective (NA ≥ 1.25)

Computer :

- Windows 10 or later
- Minimum 8 GB RAM (16 GB recommended)

Comparing With ISCN Standards



Why Choose ChromoTraQ?

Build Your System Within Your Budget

- You can reuse existing microscopes / visualizing devices / PCs with ChromoTraQ software and make your research and clinical ready karyotyping system without purchasing high end hardware systems.

Best in class AI Accuracy

- ChromoTraQ offers best accuracy in the market, in automatically separating chromosomes and classifying, making the karyotyping easy as never before.

Advanced Visualization & Comparison Tools

- ChromoTraQ contains an array of advanced image comparison features to visualize and analyze anomalies in chromosomes including Digital chromosome straightening, side by side comparison with ISCN reference ideograms together with Modern, and intuitive UI.

How It Works

Step 1 – Image Acquisition

- Capture or import metaphase spread images from your microscopy system.

Step 2 – AI Segmentation

- Automatic chromosome detection and boundary identification.

Step 3 – AI Classification & Pairing

- Chromosome classification and arrangement according to ISCN standards.

Step 4 – Smart Comparison

- Side-by-side comparison with reference ideograms and stored chromosome datasets.

Step 5 – Report Generation

- Generate structured, professional cytogenetic reports instantly.

About Quantine Technologies

Founded in 2015 and headquartered in Malabe, Sri Lanka, Quantine Technologies is a specialized software company focused on:

- Artificial Intelligence & Computer Vision
- Biomedical Software Solutions
- Scientific Imaging Systems
- Laser & Device Control Software

Our mission is to develop advanced yet practical technologies that empower research and clinical laboratories worldwide.

Contact Information:

www.quantinetech.com

info@quantinetech.com

+94 718 679 805

Scan Me

